Mathematical Studies Analysis

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Fall 2026

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1 The fundamental theorem of calculus

1.1 Differentiation of monotone functions

Theorem (Lebesgue's differentiation theorem)**.** Let $I \subset \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a monotone function. Then f is differentiable a.e. on I.

To show this, we need a few lemmas.

Lemma. Let $I \subset \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let

$$E = \left\{x \in I^\circ : \text{ there exist } f'_+(x) \text{ and } f'_-(x) \text{ and } f'_+(x) \neq f'_-(x)\right\}.$$

Then E is countable.

Proof. First observe that for any function $g : I \to \mathbb{R}$, the set of its strict local minimum points in I° is countable. Set

$$E_- = \left\{x \in E : f'_-(x) < f'_+(x)\right\}.$$

For every $r \in \mathbb{Q}$, define

$$E_r = \left\{x \in E_- : f'_-(x) < r < f'_+(x)\right\},$$

and let $g_r(t) = f(t) - rt$ for $t \in I$. By density of rationals, $E_- = \bigcup_{r\in\mathbb{Q}} E_r$. If $x \in E_r$, then

$$(g_r)'_-(x) < 0 < (g_r)'_+(x).$$

It follows that $g_r(y) > g_r(x)$ for every $y \neq x$ sufficiently close to x, i.e. x is a strict local minimum point of g_r. Therefore E_r is countable and thus E_- is countable. Applying the same argument to $-f$ completes the proof for the lemma. □

Lemma. Let $E \subset \mathbb{R}$ be bounded, and $\mathcal{F}$ be a family of open intervals such that each $x \in E$ is the left endpoint of an interval $(x, x + h_x) \in \mathcal{F}$. Then there for every $\varepsilon > 0$, there exist disjoint intervals $I_1, \ldots, I_n \in \mathcal{F}$ such that

$$\mathcal{L}^1_o\left(E \cap \bigcup_{k=1}^{n} I_k\right) \geq \mathcal{L}^1_o(E) - \varepsilon.$$

Proof. If $E = \emptyset$, take the empty collection. If $E \neq \emptyset$ and $\mathcal{L}^1_o(E) \leq \varepsilon$, any one interval in $\mathcal{F}$ suffices. Thus assume that $L^1_o(E) > \varepsilon$. For each $x \in E$, choose an interval $(x, x + h_x) \in \mathcal{F}$ and define

$$E_n = \left\{x \in E : h_x > n^{-1}\right\}.$$

Note that $E_n \subset E_{n+1}$ and $E = \bigcup_{n=1}^{\infty} E_n$. By coninuity from below of Lebesgue outer measure (exercise), we have $\mathcal{L}^1_o(E_n) \to \mathcal{L}^1_o(E)$. Hence we can find m large enough such that $\mathcal{L}^1_o(E_m) > \mathcal{L}^1_o(E) - \frac{\varepsilon}{2}$. For each $k \in \mathbb{N}$, let $\eta_k = \varepsilon 2^{-k-1}$, then $\sum_{k=1}^{\infty} \eta_k = \frac{\varepsilon}{2}$.

Let $a_1 = \inf E_m$. By definition of infimum we can find $x_1 \in E_m$ with $a_1 \leq x_1 < a_1 + \eta_1$, and correspondingly $I_1 = (x_1 + h_1) \in \mathcal{F}$ with $h_1 > m^{-1}$. Suppose $I_1, \ldots, I_k$ has been chosen.

If $\{x \in E_m : x \geq x_k + h_k\} = \emptyset$, we stop. Otherwise let $a_{k+1} = \inf\{x \in E_m : x \geq x_k + h_k\}$. Then there is $x_{k+1} \in E_m$ such that $a_{k+1} \leq x_{k+1} \leq a_{k+1} + \eta_{k+1}$ and correspondingly $I_{k+1} = (x_{k+1}, x_{k+1} + h_{k+1}) \in \mathcal{F}$ with $h_{k+1} > m^{-1}$. Note that

$$x_{k+1} \geq x_k + h_k > x_k + m^{-1},$$

and $E_m \subset E$ is bounded, the construction must stop after finitely many steps. Moreover, $I_1, \ldots, I_n$ are disjoint by construction.

Now let

$$S = \bigcup_{k=1}^{n} I_n, \quad T = \bigcup_{k=1}^{n} (x_k - \eta_k, x_k].$$

The points of E_m skipped at the k-th step belong to $[a_k, x_k] \subset (x_k - \eta_k, x_k]$. Thus $E_m \subset S \cup T$, then

$$\begin{aligned} \mathcal{L}^1_o(E) - \frac{\varepsilon}{2} &< \mathcal{L}^1_o(E_m) \leq \mathcal{L}^1_o(E_m \cap S) + \mathcal{L}^1_o(E_m \cap T) \\ &\leq \mathcal{L}^1_o(E_m \cap S) + \sum_{k=1}^{n} \eta_k \leq \mathcal{L}^1_o(E_m \cap S) + \frac{\varepsilon}{2}. \end{aligned}$$

This concludes the proof. □

Lemma. Let $E \subset \mathbb{R}$ be bounded, $\mathcal{F}$ be family of open intervals such that for all $x \in E$ and $\delta > 0$, there exists $(x, x + h_x) \in \mathcal{F}$ with $0 \leq h_x < \delta$. Then for all $\varepsilon > 0$, there exists disjoint intervals $I_1, \ldots, I_m \in \mathcal{F}$ such that

$$\mathcal{L}^1_o\left(E \cap \bigcup_{i=1}^{m} I_i\right) \geq \mathcal{L}^1_o(E) - \varepsilon, \quad \sum_{i=1}^{m} \mathcal{L}^1(I_i) \leq \mathcal{L}^1_o(E) + \varepsilon.$$

Proof. By the definiton of Lebesgue outer measure, find an bounded open set $U \supset E$ such that $\mathcal{L}^1_o(U) \leq \mathcal{L}^1_o(E) + \varepsilon$. Let $\mathcal{F}'$ be the subfamily of intervals $(x, x + h_x) \in \mathcal{F}$ contained in U. Note that for each $x \in E \subset U$, there must exists at least one such interval, since U contains a ball centered at x and there are intervals of arbitrarily small length.

Apply the previous lemma to the family $\mathcal{F}'$ to find disjoint intervals $I_1, \ldots, I_n \subset \mathcal{F}'$ such that

$$\mathcal{L}^1_o\left(E \cap \bigcup_{k=1}^{n} I_k\right) \geq \mathcal{L}^1_o(E) - \varepsilon.$$

Since the intervals are disjoint and contined in U, it follows that

$$\sum_{k=1}^{n} \mathcal{L}^1(I_k) \leq \mathcal{L}^1_o(U) \leq \mathcal{L}^1_o(E) + \varepsilon.$$

The proof is then complete. □

Definition. For a function $f : E \to \mathbb{R}$, define

$$\liminf_{x\to x_0} f(x) = \sup_{r>0} \inf_{0<d(x,x_0)<r} f(x), \quad \limsup_{x\to x_0} f(x) = \inf_{r>0} \sup_{0<d(x,x_0)<r} f(x),$$

Exercise. $\liminf_{x\to x_0} f(x) = l \in \mathbb{R}$ iff for every $\varepsilon > 0$:

1. there exists $\delta > 0$ such that $l - \varepsilon < f(x)$ for all $x \in E$ with $0 < d(x, x_0) < \delta$.
2. For every $\delta > 0$, there exists $x \in E$ with $0 < d(x, x_0) < \delta$ such that $f(x) < l + \varepsilon$.

Also, $\liminf_{x\to x_0} f(x) = -\infty$ iff for every $M > 0$ and $\delta > 0$, there exists $x \in E$ with $0 < d(x, x_0) < \delta$ such that $f(x) < -M$.

Now we are ready to show Lebesgue's differentiation theorem. Recall the statement of the theorem below:

Theorem (Lebesgue's differentiation theorem)**.** Let $I \subset \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a monotone function. Then f is differentiable a.e. on I.

Proof. Define **Dini's derivatives** as follows:

$$\overline{D}_+ f(x_0) = \limsup_{x\to x_0^+} \frac{f(x) - f(x_0)}{x - x_0}, \qquad \underline{D}_+ f(x_0) = \liminf_{x\to x_0^+} \frac{f(x) - f(x_0)}{x - x_0},$$
$$\overline{D}_- f(x_0) = \limsup_{x\to x_0^-} \frac{f(x) - f(x_0)}{x - x_0}, \qquad \underline{D}_- f(x_0) = \liminf_{x\to x_0^-} \frac{f(x) - f(x_0)}{x - x_0}.$$

WLOG assume f is increasing, and assume $I = [a, b]$. Let

$$\begin{aligned} E &= \left\{x \in I^\circ : \underline{D}_+ f(x) < \overline{D}_+ f(x)\right\} \\ &= \bigcup_{\substack{r,s\in\mathbb{Q} \\ 0<r<s}} \left\{x \in I^\circ : \underline{D}_+ f(x) < r < s < \overline{D}_+ f(x)\right\} \\ &= \bigcup_{\substack{r,s\in\mathbb{Q} \\ 0<r<s}} E_{r,s} \end{aligned}$$

It suffices to show that $\mathcal{L}^1(E_{r,s}) = 0$. Let $\varepsilon > 0$ be given.

Let $x \in E_{r,s}$. We have

$$\liminf_{h\to 0^+} \frac{f(x+h) - f(x)}{h} < r.$$

Then for each $\delta > 0$, there exists $0 < h_x < \delta$ such that $h_x^{-1}(f(x + h_x) - f(x)) < r$. Let $\mathcal{F}$ be the collection of $(x, x + h_x)$, where h_x is selected as aforementioned while δ ranges over all positive

value, and x ranges over $E_{r,s}$. It is then clear that $\mathcal{F}$ satisfies the condition for the previous lemma.

Apply previous lemma, there exists $I_1, \ldots, I_m \in \mathcal{F}$ disjoint such that

$$\mathcal{L}^1_o\left(E_{r,s} \cap \bigcup_{i=1}^{m} I_i\right) \geq \mathcal{L}^1_o(E_{r,s}) - \varepsilon, \tag{1}$$

$$\sum_{i=1}^{m} \mathcal{L}^1(I_i) \leq \mathcal{L}^1_o(E_{r,s}) + \varepsilon. \tag{2}$$

Write $I_i = (x_i, x_i + h_i)$, then

$$\sum_{i=1}^{n} f(x_i + h_i) - f(x_i) < r \sum_{i=1}^{n} h_i \leq r\mathcal{L}^1_o(E_{r,s}) + r\varepsilon. \tag{3}$$

Now let $V = \bigcup_{i=1}^{m}(x_i, x_i + h_i)$, which is open. If $x \in E_{r,s} \cap V$, we have

$$s < \limsup_{t \to 0} \frac{f(x+t) - f(x)}{t}.$$

Then for every $\delta > 0$, there exists f_x such that $0 < t_x < \delta$ and $s < t_x^{-1}(f(x + t_x) - f(x))$. Define $\mathcal{G}$ to be the collection of $(x, x + t_x)$ that is contained in V, where t_x is selected as aforementioned while $\delta > 0$ ranges over all positive value, and x ranges over $E_{r,s} \cap V$.

The previous lemma again gives $J_1, \ldots, J_l \in \mathcal{G}$ disjoint such that

$$\mathcal{L}^1_o\left(E_{r,s} \cap V \cap \bigcup_{k=1}^{l} J_k\right) \geq \mathcal{L}^1_o(E_{r,s} \cap V) - \varepsilon.$$

Write $J_k = (y_k, y_k + t_k)$, we have

$$\begin{aligned} \sum_{k=1}^{l} f(y_k + t_k) - f(y_k) &\geq s \sum_{k=1}^{l} t_k \\ &\geq s\mathcal{L}^1_o(E_{r,s} \cap V) - s\varepsilon \\ &\geq s\mathcal{L}^1_o(E_{r,s}) - 2s\varepsilon. \end{aligned} \tag{4}$$

Now compare (3) and (4). Since f is increasing and each J_k lies in one of the interval I_i, it is not hard to see that

$$\sum_{i=1}^{n} f(x_i + h_i) \geq \sum_{k=1}^{l} f(y_k + t_k) - f(y_k).$$

It follows that

$$\mathcal{L}^1_o(E_{r,s}) \leq \frac{2s + r}{s - r}\varepsilon.$$

Letting $\varepsilon \to 0$ completes the proof for $\mathcal{L}^1(E_{r,s}) = 0$. This implies that $f'_+(x)$ exists for a.e. $x \in [a,b]$. Similarly, $f'_-(x)$ exists for a.e. $x \in [a,b]$. A previous lemma shows that the left and right derivatives are equal a.e. Therefore, $f'(x)$ exists a.e.

It remains to show the left and right derivatives are a.e. finite. Define

$$g(x) = \begin{cases} f(x), & x \in [a,b] \\ f(b), & x \geq b. \end{cases}$$

Define also

$$g_n(x) = \frac{g(x+n^{-1}) - g(x)}{n^{-1}}$$

for $x \in [a,b]$ and $n \in \mathbb{N}$. Then, $g_m(x) \to f'(x)$ for a.e. $x \in [a,b]$. Since f is monotone, it is measurable and thus g_m is also measurable. By Fatou's lemma,

$$\begin{aligned}
\int_a^b f'(x)\,dx &\leq \liminf_{n\to\infty} \int_a^b g_n(x)\,dx \\
&= \liminf_{n\to\infty} n\int_a^b g(x+n^{-1})\,dx - n\int_a^b g(x)\,dx \\
&= \liminf_{n\to\infty} n\int_a^{b-n^{-1}} g(x+n^{-1})\,dx + n\int_{b-n^{-1}}^b f(b) - n\int_a^b g(x)\,dx \\
&= \liminf_{n\to\infty} n\int_{a+n^{-1}}^b g(t)\,dt + f(b) - n\int_a^b g(t)\,dt \\
&= \liminf_{n\to\infty} f(b) - \int_a^{a+n^{-1}} f(t)\,dt \\
&\leq f(b) - f(a).
\end{aligned}$$

Therefore, f' is integrable, which implies $\{f' = \infty\}$ has measure zero.

For a general interval $I \subset \mathbb{R}$, I can be written as a countable union of bounded interval, the proof for the theorem is complete. □

Remark. From the proof above, we see that for monotone function $f : [a,b] \to \mathbb{R}$, we have

$$\int_a^b f' \leq f(b) - f(a).$$

There exists a monotone function for which inequality is strict. Indeed, let $f : [0,1] \to [0,1]$ be the Cantor function. Then, $f' = 0$ a.e. but $f(1) = 1$ and $f(0) = 0$.

1.2 Differentiation of BV functions

Definition. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}^N$. The **pointwise variation** of f on the interval I is defined as

$$\operatorname{Var} f = \sup \left\{ \sum_{i=1}^{n} \|f(x_i) - f(x_{i-1})\| \right\},$$

where the supremum is taken over all partition $P = \{x_0, \ldots, x_n\}$ of I, $n \in \mathbb{N}$. A function has **bounded pointwise variation** if $\operatorname{Var} f < \infty$.

The space of all functions $f : I \to \mathbb{R}^N$ with bounded pointwise variation is denoted by $BV(I; \mathbb{R}^N)$.